

MediLink Interface Design

Module Code: COMP50070

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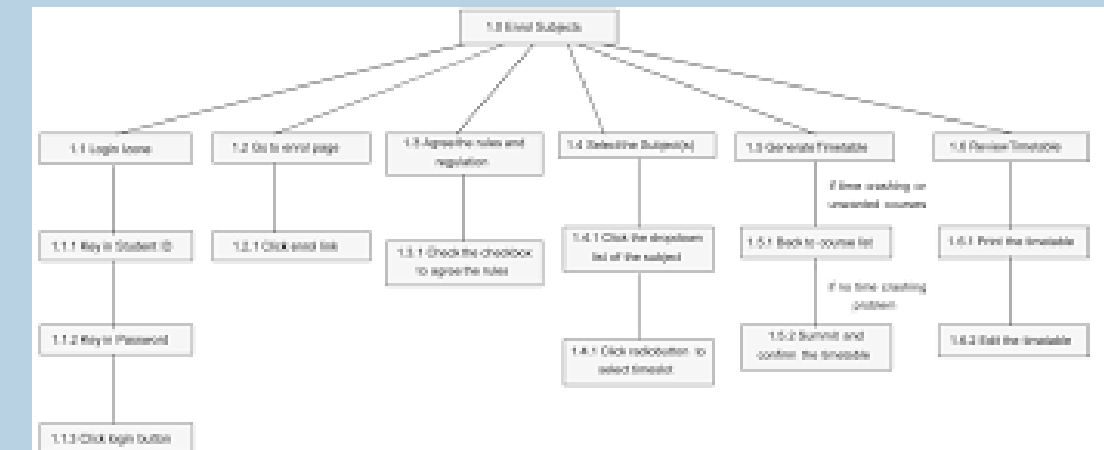


Scinario Overview

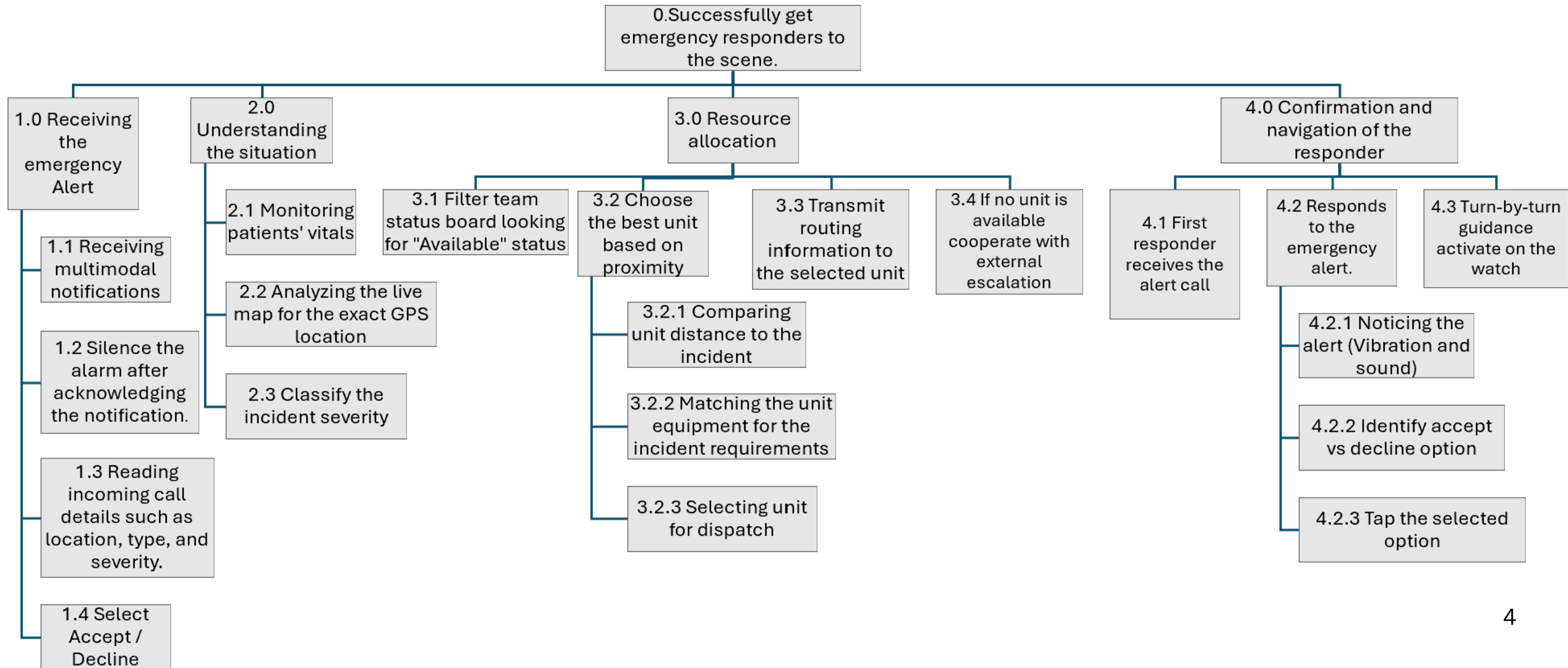
- MediLink is a system used by a hospital's emergency response team.
- Used in
 1. Desktop - Dispatcher (Monitoring & Coordination)
 2. Smartwatch - Paramedic (Field Response)
- Users operate devices in different environments.

HTA DIAGRAM

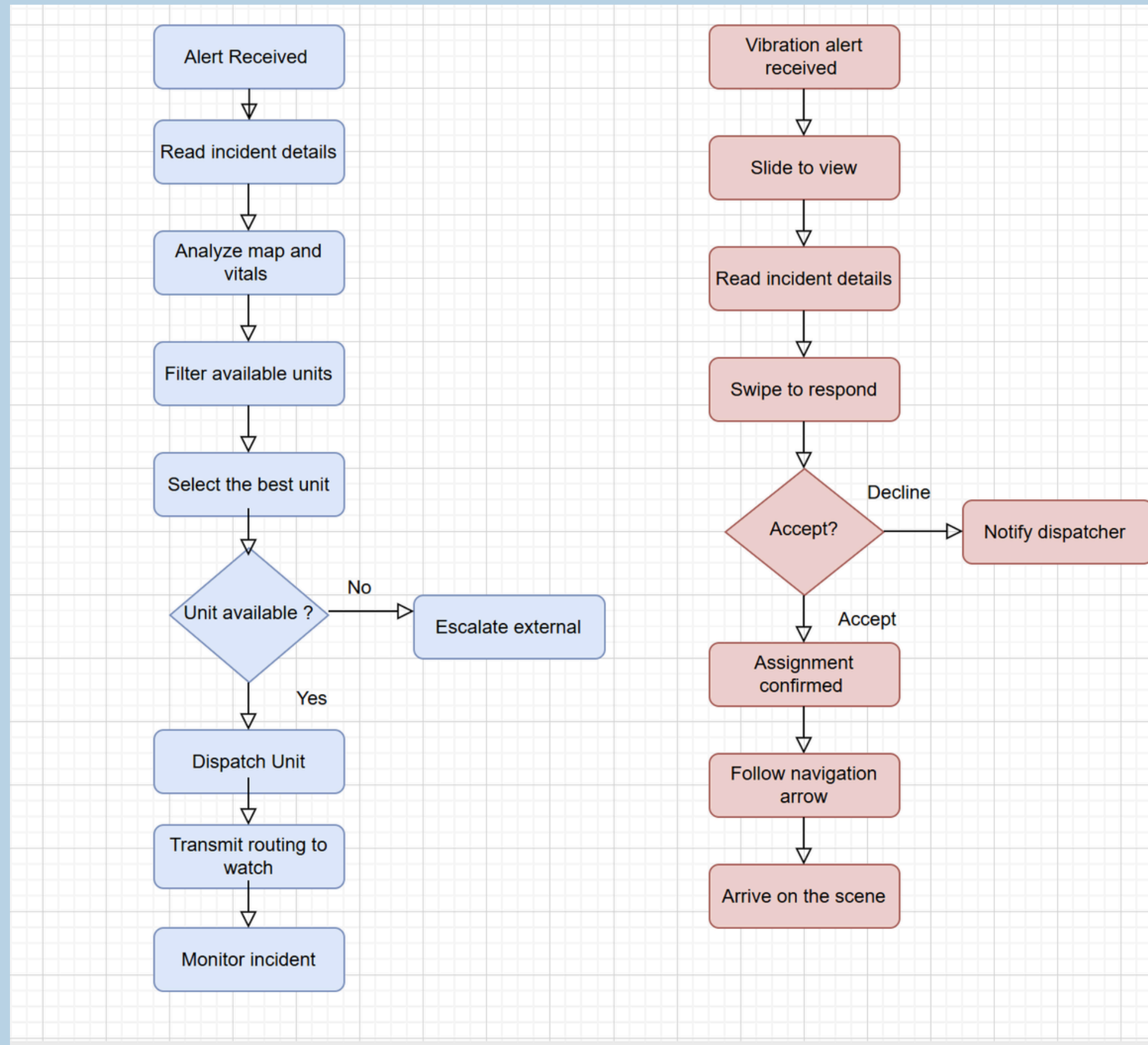
- HTA is a systematic method used by UX designers to break down complex tasks into simpler subtasks.
- Dispatcher:
 1. Complex, tasks contain multiple steps
 2. Focused on monitoring, decision making & coordination.
- Paramedic:
 1. Fast, minimal steps
 2. Act quickly in a critical situation



MediLink HTA Diagram



Process Flow

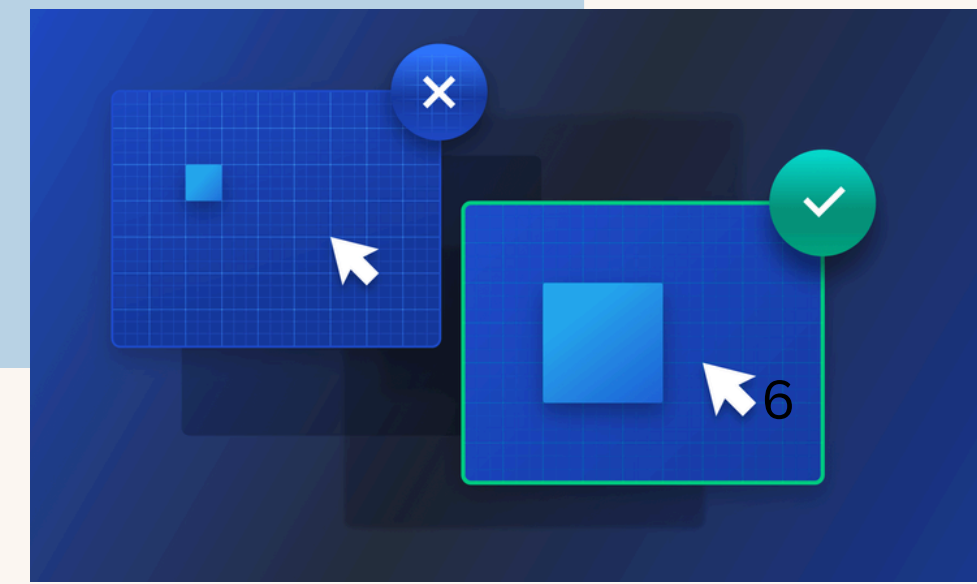


Fitt's Law

The movement time is dependent on the size of the target and the distance (MacKenzie, 1992).

Formula

$$MT = a + b \cdot \log_2(2D/W)$$



Desktop

- The mouse is used as the input device.
- Used small to medium targets.
- **Examples -**
Unit cards: 245 x 99px
Dispatch button: 245 x 99px
- Faster selection due to larger targets.

Smart Watch

- Finger input
- Large touch targets are required.
- **Examples -**
Accept/ Decline: 113 x 36px
- Less MT due to larger buttons.
- Faster interaction and fewer errors.

Hick's Law

Response time increases with the number of available options (Hick, 1952).

Formula

$$RT = b \cdot \log_2(N+1)$$



Desktop

- The dispatcher can accept more options.
- Supports complex decisions.
- Examples: Multiple units, Filters

Smart Watch

- Needed fewer choices.
- Designed for quick decisions.
- Example: Accept / Decline

Desktop Wirefrme

MedLink

Active emergency

Dispatcher

Shift

System Status

Current time

TEAM STATUS

Filter

INCIDENT TYPE

LOCATION

SEVERITY

SELECTED UNIT

Dispatch Unit

Edit Assignment

Escalate External

PATIENT VITALS

HEART RATE

SP02

BP

RESP RATE

INCIDENT LOG

Time

Incident

ROUTING INFO

Desktop Wirefrme

MedLink

Active emergency

Dispatcher : A Perera

Shift 08:00-20:00

System Online

14:32:07

TEAM STATUS

Filter : Available

AMB-04

Available

1.2 km . ALS Equipped

ETA : 4min Crew: 2

AMB-07

Available

3.8 km . BLS Equipped

ETA : 11min Crew: 2

AMB-02

En Route

Dispatch 14.28

AMB-11

On Scene

Main Road, Colombo

EXTERNAL ESCALATION



INCIDENT TYPE	LOCATION	SEVERITY	SELECTED UNIT
Cardiac Arrest	Block 4-B	High	AMB - 04

Dispatched

Edit Assignment

Escalate External

PATIENT VITALS

HEART RATE

112 bpm

SP02

94%

BP

140 / 190

RESP RATE

22 / min

INCIDENT LOG

Time	Incident
14:31	Emergency call received. Cardiac arrest.
14:32	Severity: Critical (Level 3)
14:32	AMB-04 selected

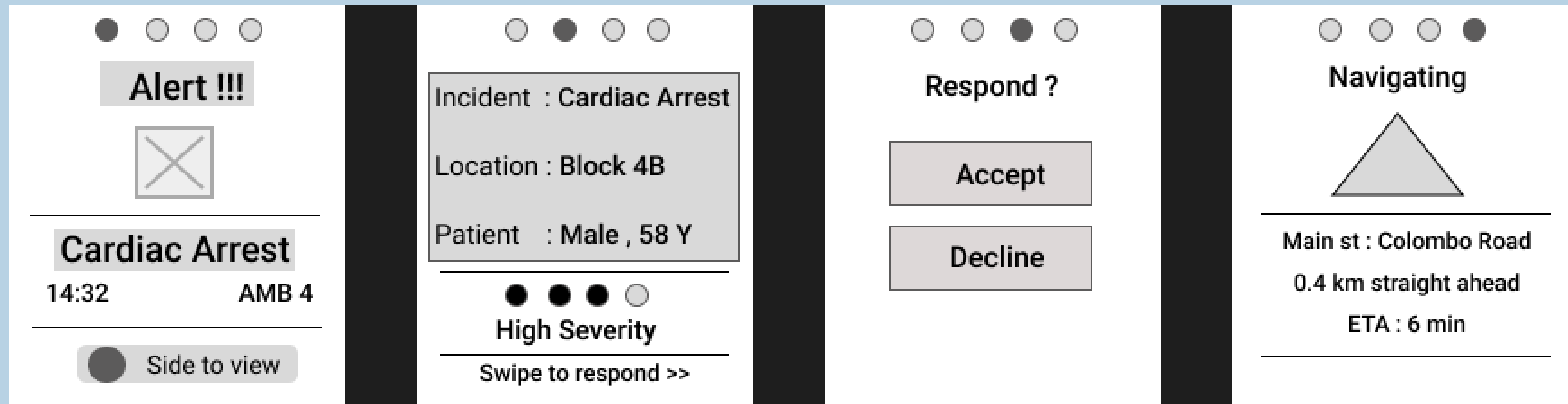
ROUTING INFO

Route - Main street , Colombo

ETA : 6 min

Distance : 2.1 km

Smartwatch Wireframes



Screen 1:
Alert screen

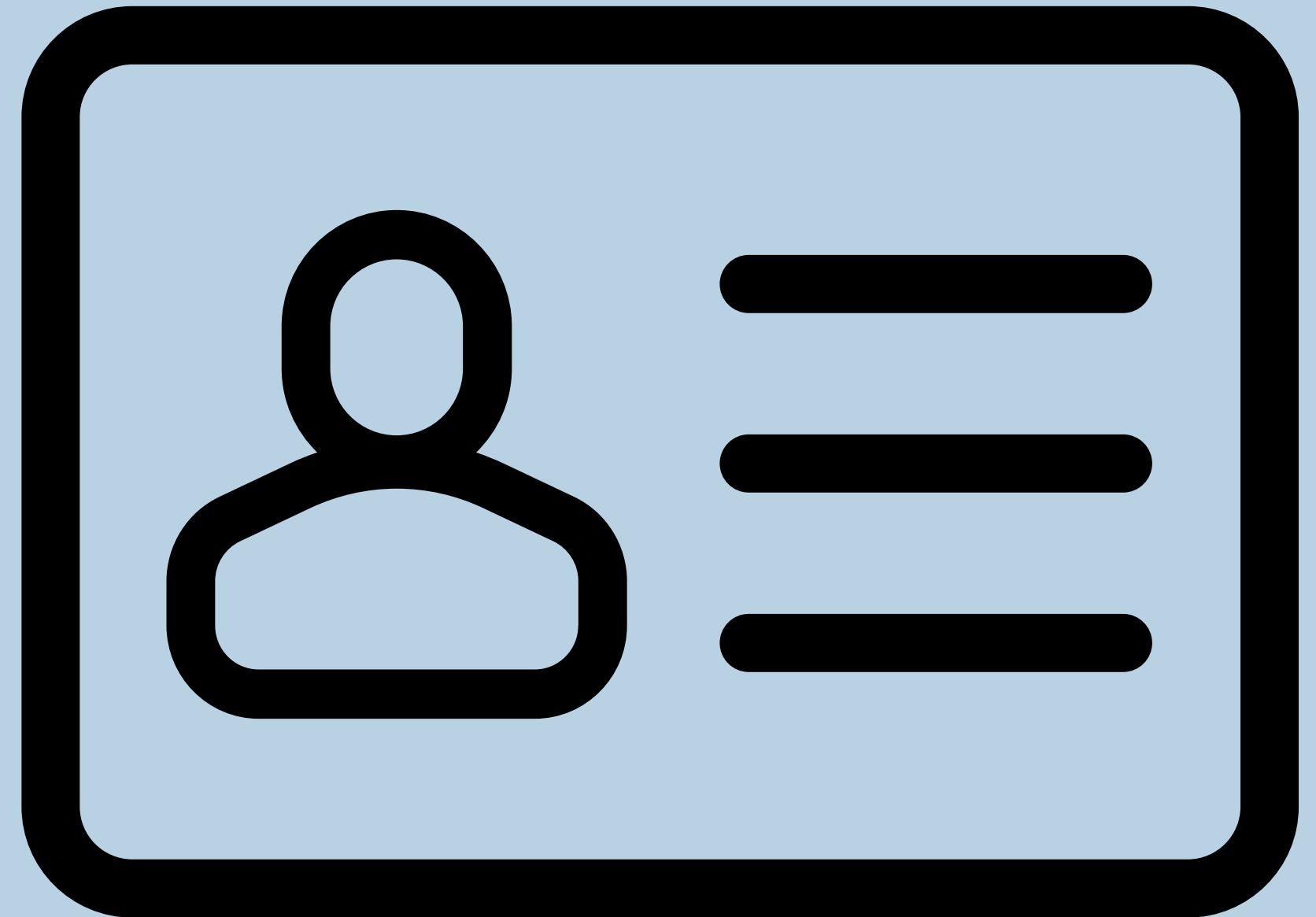
Screen 2:
Incident
screen

Screen 3:
Respond
screen

Screen 4:
Navigation
screen

Persona

- Personas are used to represent different user contexts.



Dispatcher's Persona



Chelsa Fernando

City General Hospital - Control Center

**Role - Emergency Dispatch
Coordinator**

“I need everything in front of me at once. Switching screens cost seconds and seconds cost lives.”

Goals & Needs

- Monitor all incident through a single dashboard
- Simultaneously monitor ambulance locations & patient vitals.
- Make decisions within or under 30 seconds
- Responding using quick mouse actions
- Access Traffic & ETA without switching screens

Environmental Constrains

- Control room with dim lights
- Dual 27 inch monitor setup
- Seated, stationary posture
- Input from keyboard & mouse
- Background radio noise
- Handling 3-5 calls at once

Cognitive & physical stress

COGNITIVE LOAD HIGH



MOTOR PRECISION HIGH



PHYSICAL STRESS MED



TIME PRESSURE HIGH



Pain points & Frustration

- Low contrast text lead to eye strain overtime
- Cluttered UI makes it difficult to maintain focus
- Switching screens between screens lead to decision delays

Tools & Tech

CAD System

Keyboard
Shortcut

Live Map Feed

Radio Console

Patient vitals
monitor

Paramedic's Persona



Sam Peris

City General Hospital - Control Center

Role - Senior Paramedic

“ I can’t stop moving to read a screen. I need to know : go or no-go. That’s it. ”

Goals & Needs

- Accept or decline an emergency dispatch with a single wrist interaction.
- Receive navigation prompts on watch screen to support quick 1-2 second glance viewing
- Patient information is limited to 3 key points per screen.

Environmental Constrains

- In motion
- Wearing latex gloves
- Bright outdoor sunlight
- Wrist only interaction
- Adrenaline spike- tremor
- Rain conditions

Tools & Tech

- Smart watch
- Ambulance MDT
- Haptic Alerts

Cognitive & Physical Profile

- Cognitive Load Max
- Motor Precision Low
- Physical stress High
- Time pressure Max

Pain Points & Frustration

- Small buttons cause mis- taps, especially when wearing gloves or moving quickly.
- Too many steps in menus slow down decisions under pressure.
- Low screen visibility in bright sunlight makes information hard to read.

Conclusion

- Different interface designs in the same system.
- Fitt's Law - Speed and accuracy are dependent on the target size.
- Hick's Law - Fewer options lead to faster decisions.
- Smartwatch and Desktop wireframes have been built according to the Fitts' Law and Hick's Law.
- Design must match the real-world context.

Figma Links

- Desktop wireframes

<https://www.figma.com/design/tSe5g9eHH9a9kln8wZntQJ/Wireframes?node-id=0-1>

- Smartwatch wireframes

<https://www.figma.com/design/tSe5g9eHH9a9kln8wZntQJ/Wireframes?node-id=39-224>

References

1. Hick, W.E., 1952. On the rate of gain of information. Quarterly Journal of Experimental Psychology, 4(1), pp.11–26.
2. Mackay, C.J. (1980) 'The measurement of mood and psychophysiological activity using self-report techniques', in Gale, A. and Christie, B. (eds.) Psychophysiology and the Electronic Workplace. Chichester: Wiley, pp. 501–517.

Thank
you !!!

